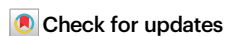


YTHDC1 functions as a molecular chaperone to suppress ALS-linked hnRNPA1 mutants from aggregation

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Proteostasis failure drives multiple neurodegenerative disorders (NDs), and ATP-independent chaperone pathways that support neuronal proteostasis remain poorly defined. Here, we identify the N⁶-methyladenosine (m⁶A)-binding protein YTHDC1 as an ATP-independent molecular chaperone, whose activity is mediated by a highly acidic polyaspartate/glutamate (polyD/E) segment. YTHDC1 prevents protein misfolding and aggregation, unfolds kinetically trapped substrates, and resolubilizes pre-formed aggregates. Deletion of the polyD/E segment abolishes these activities, whereas aromatic-cage mutants retain chaperone activity, demonstrating independence from m⁶A recognition. We identify the amyotrophic lateral sclerosis (ALS)-associated RNA-binding protein hnRNPA1 as a YTHDC1 client. YTHDC1 maintains liquid-like hnRNPA1 condensates, delays fibrillization of disease-associated mutants, and limits stress-granule sequestration, while mitigating mutant hnRNPA1-induced neurite growth defects in primary neurons. These findings define a proteostatic function of YTHDC1 and highlight its chaperone activity as a potential target for mitigating protein aggregation in ALS-related NDs.

To maintain protein homeostasis and ensure normal cellular functions, organisms have evolved sophisticated protein quality control (PQC) systems that prevent protein misfolding and aggregation, dissolve pre-existing aggregates, and facilitate their degradation^{1,2}. Classical PQC modules include heat shock protein (HSP) families, the ubiquitin-proteasome system, and lysosomal degradation pathways³. However, reliance solely on these pathways can be insufficient to stabilize the myriad proteomes across cellular contexts, motivating the search for additional PQC mechanisms. Recent work has highlighted the significant roles of atypical, ATP-independent molecular chaperones such as DAXX, TRIM11, and Kap β 2 that preserve the stability of key proteins, including p53, tau and FUS^{4–6}. Of particular interest, DAXX is a

polyaspartate/glutamate (polyD/E) protein that bears an extensively negatively charged poly-Asp/Glu region and functions as a molecular chaperone, disaggregase, and unfoldase in an ATP-independent manner, suggesting that densely negatively charged amino acids may drive chaperone activity.

As one of the most abundant RNA modifications in cells, m⁶A broadly plays a critical role in RNA processing and metabolism⁷. The YTH family of m⁶A-selective “reader” proteins mediates the effects of methylation modification on the fates of eukaryotic mRNA. YTHDC1, a YTH family member, promotes nuclear export of methylated mRNAs and is essential for diverse cellular processes⁸. Structurally, YTHDC1 contains a YTH domain flanked by intrinsically disordered regions

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(IDRs) at both termini⁹. The N-terminal IDR of YTHDC1 harbors a highly charged polyD/E tract reminiscent of DAXX's chaperone module⁴. These observations led us to hypothesize that YTHDC1 may possess PQC-related functions.

The hnRNPA1 is a multifunctional RBP involved in RNA splicing, transcriptional regulation, and RNA transport, thereby maintaining mRNA processing and cellular homeostasis¹⁰. Mutations, hnRNPA1^{D262N} and hnRNPA1^{D262V}, in hnRNPA1 have been linked to neurodegeneration, particularly amyotrophic lateral sclerosis (ALS)¹¹. Familial ALS cases have identified pathogenic mutations in the prion-like domain (PrLD) of hnRNPA1 associated with inherited degeneration of muscle, brain, motor neurons, and bone^{11,12}. These variants promote the formation of toxic protein aggregates and misfolded intermediates that drive neuronal dysfunction and degeneration^{13,14}. Here, we identified YTHDC1 as a polyD/E-containing molecular chaperone that suppresses the aggregation formation of mutant hnRNPA1 and mitigates the neurite growth defects.

Results

YTHDC1 functions as an effective molecular chaperone in vitro

Chaperones exhibit holdase activity, unfoldase activity or disaggregate activity for model substrates and disease-associated proteins^{15–19}. Since we found that YTHDC1 harbors a highly charged polyD/E tract in its N-terminal IDR, we investigated the potential molecular chaperone activity of YTHDC1. We purified full-length recombinant YTHDC1 from HEK293F cells using dual MBP/His tags (Supplementary Fig. 1a, b) and evaluated chaperone function with the canonical chaperone substrate luciferase¹⁷. Purified MBP protein served as a negative control, and HSP70/ HSP40 or HSP104^{AS035}, a potentiated HSP104 variant, served as a positive control in the presence of ATP-Mg²⁺ and an ATP-regeneration system (Supplementary Fig. 1c, d)^{3,16}. To assess the holdase activity, luciferase was incubated at 42 °C with YTHDC1 and assayed by luminescence. Luciferase with MBP inactivated rapidly after heat incubation, whereas recombinant YTHDC1 preserved luciferase activity akin to HSP70-40 (Fig. 1a and Supplementary Fig. 1e). Turbidity measurement at OD₆₀₀ showed that YTHDC1 efficiently prevents the aggregate formation, comparable to HSP70-40 (Fig. 1b).

To test unfoldase activity of YTHDC1, we employed LucDH56 (LucD), a monomeric luciferase variant misfolded by freeze-thaw (Supplementary Fig. 1f, g). This misfolded LucD displayed increased thioflavin T (ThT) binding relative to native protein, indicating elevated β -sheet content^{4,16}. Co-incubation with YTHDC1 markedly reduced ThT fluorescence signals (Fig. 1c). Trypsin sensitivity assay showed that YTHDC1 transiently increased LucD sensitivity to cleavage at early time points, followed by reduced sensitivity upon prolonged incubation, indicating a rapid unfolding of the substrate and a subsequent refolding (Fig. 1d, e).

To examine the disaggregate activity of YTHDC1, heat-denatured luciferase was pelleted, resuspended and co-incubated with YTHDC1. Sedimentation assays demonstrated time- and dose-dependent resolubilization from the pellet (PE) to supernatant (SN) (Fig. 1f, g), accompanied by recovery of enzymatic activity approaching HSP104-mediated levels (Fig. 1h and Supplementary Fig. 1h). Furthermore, substrate titrations revealed that YTHDC1 facilitated the reactivation following saturation kinetics (Fig. 1i). Together, YTHDC1 functions as a molecular chaperone, prevents aggregation, unfolds misfolded substrates, and reactivates misfolded proteins in vitro.

YTHDC1 suppresses intracellular protein aggregation and oligomerization

Next, we evaluated the influence of YTHDC1 on intracellular misfolded proteins. In HEK293T cells co-expressing YTHDC1 and the unstable luciferase derivative Luc-DM (N-terminal NLS, C-terminal GFP)¹⁷, the enzymatic activity of Luc-DM increased within 12 h (Fig. 1j).

Fractionation revealed a marked reduction in NP-40-insoluble Luc-DM (Fig. 1k). Quantitative analysis further demonstrated that Luc-DM mRNA levels remained unchanged, indicating that YTHDC1 reduces insoluble Luc-DM through a post-transcriptional mechanism rather than by altering its transcription (Fig. S1i). To extend these findings, we depleted YTHDC1 in HeLa cells and stained intracellular aggregates using PROTEOSTAT²⁰. YTHDC1 knockdown led to a pronounced accumulation of aggregates (Fig. 1l, m and Supplementary Fig. 2a), demonstrating that YTHDC1 potently suppresses the formation of intracellular protein aggregates.

We also observed that knockdown of YTHDC1 in cells resulted in a significant increase in oligomer levels detected by A11 antibody (Supplementary Fig. 2b, c). To further probe oligomerization, we used bimolecular fluorescence complementation (BiFC) system with α -synuclein (α -Syn), fused to Venus fragments (VN-S and S-VC) (Supplementary Fig. 2d)^{21,22}. YTHDC1 expression markedly decreased BiFC fluorescence (Fig. 1n, o), while the protein levels of VN-S and S-VC remained unchanged (Supplementary Fig. 2e), indicating suppression of substrate oligomerization by YTHDC1 at the cellular level.

Chaperone activity is independent of YTHDC1's m⁶A reader function

Because YTHDC1 is a well-characterized m⁶A reader^{22,23}, we asked whether its chaperone activity is associated with m⁶A recognition. YTHDC1 contains a YTH domain recognizing m⁶A modification flanked by IDRs at both termini (Fig. 1p). Mutations disrupting the YTH aromatic cage (W377A, YTHDC1^W; W377A/W428A, YTHDC1^{WW}) abolish m⁶A binding while preserving overall fold²⁴ (Fig. 1q). Both YTHDC1^W and YTHDC1^{WW} retained their ability to reduce Luc-DM aggregate formation in the cell pellets (Fig. 1r) and enhanced Luc-DM activity (Fig. 1s) comparable to wild-type YTHDC1. Together, these findings indicate that the chaperone function of YTHDC1 is not affected by deficiencies in m⁶A recognition, suggesting that the chaperone activity operates independently of m⁶A reading.

PolyD/E region is necessary for the chaperone function of YTHDC1

The N-terminal IDR of YTHDC1 contains a polyD/E tract (Fig. 1p), which has been implicated in electrostatic interactions and might contribute to chaperone activity^{4,22}. To investigate the functional domains associated with the chaperone activity of YTHDC1, we generated a polyD/E deletion mutant, YTHDC1 ^{Δ D/E} (Fig. 1q). Deleting the polyD/E region abolished suppression of Luc-DM aggregation in cells (Fig. 2a), failed to restore the enzymatic activity of Luc-DM (Fig. 2b) and eliminated inhibition of α -Syn oligomerization in BiFC (Fig. 2c, d and Supplementary Fig. 2f), demonstrating that polyD/E region is essential for cellular chaperone activity of YTHDC1.

Consistent with cellular assays, purified recombinant YTHDC1 ^{Δ D/E} protein (Fig. S2g) lost its capacity to prevent heat-induced inactivation and aggregation of luciferase (Fig. 2e, f and Supplementary Fig. 2h), failed to resolubilize and reactivate heat-denatured luciferase (Fig. 2g and Supplementary Fig. 2i). The unfoldase activity was also impaired, as evidenced by reduced ThT binding and trypsin-sensitivity (Fig. 2h, i). Furthermore, thermal denaturation shift assays revealed that luciferase incubated with YTHDC1 exhibited an increased transition temperature (T_m) compared to MBP or YTHDC1 ^{Δ D/E} (Fig. 2j). This finding indicates that the depletion of polyD/E abolishes YTHDC1's capacity to thermally stabilize its substrate. Collectively, these findings suggest that the chaperone function of YTHDC1 operates closely associated with its polyD/E repeats.

The polyD/E region promotes YTHDC1 self-stability and condensate fluidity

Depletion of polyD/E repeats abolishes YTHDC1's ability to stabilize substrates, when we performed the thermal shift assay, we also

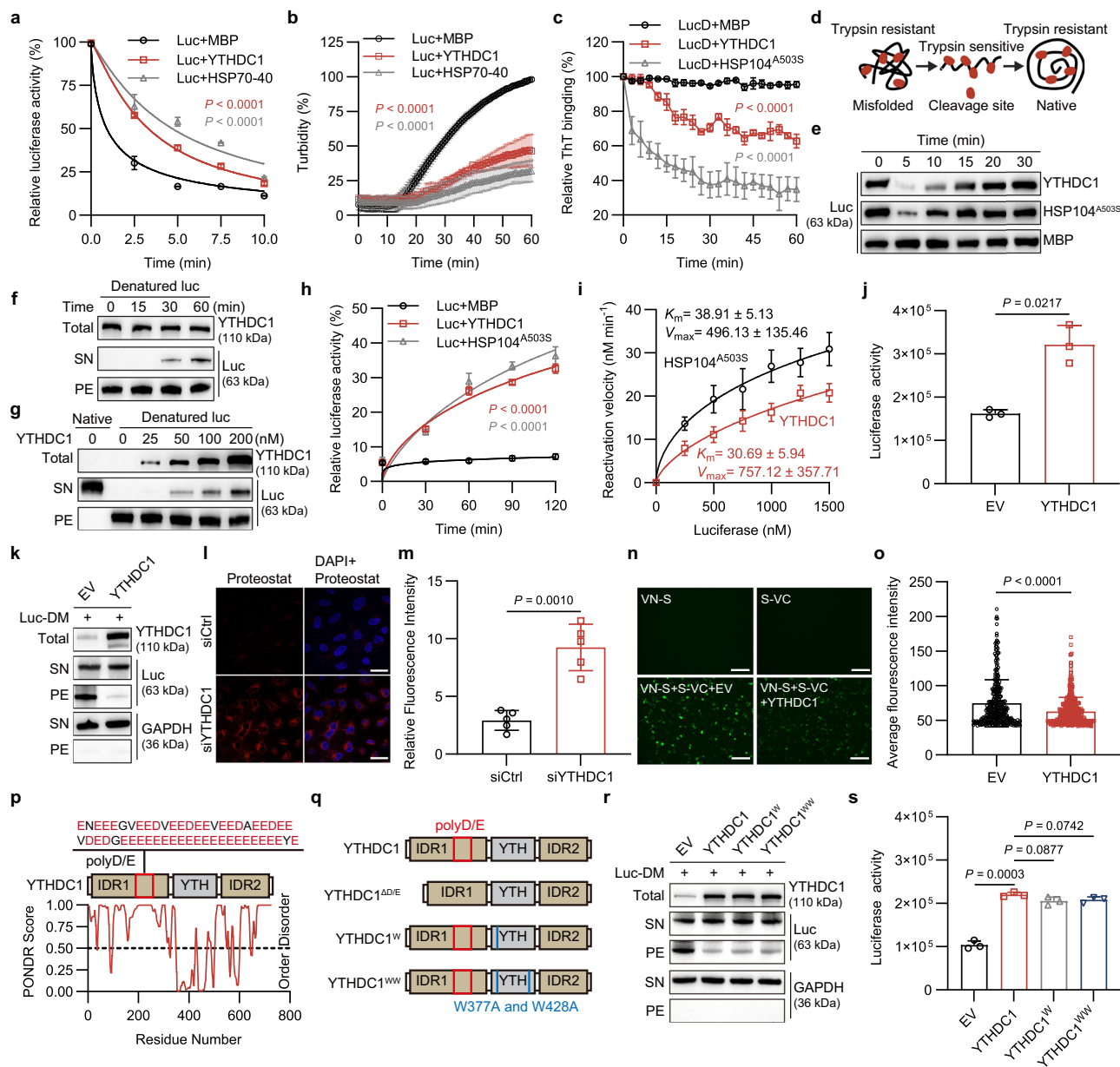


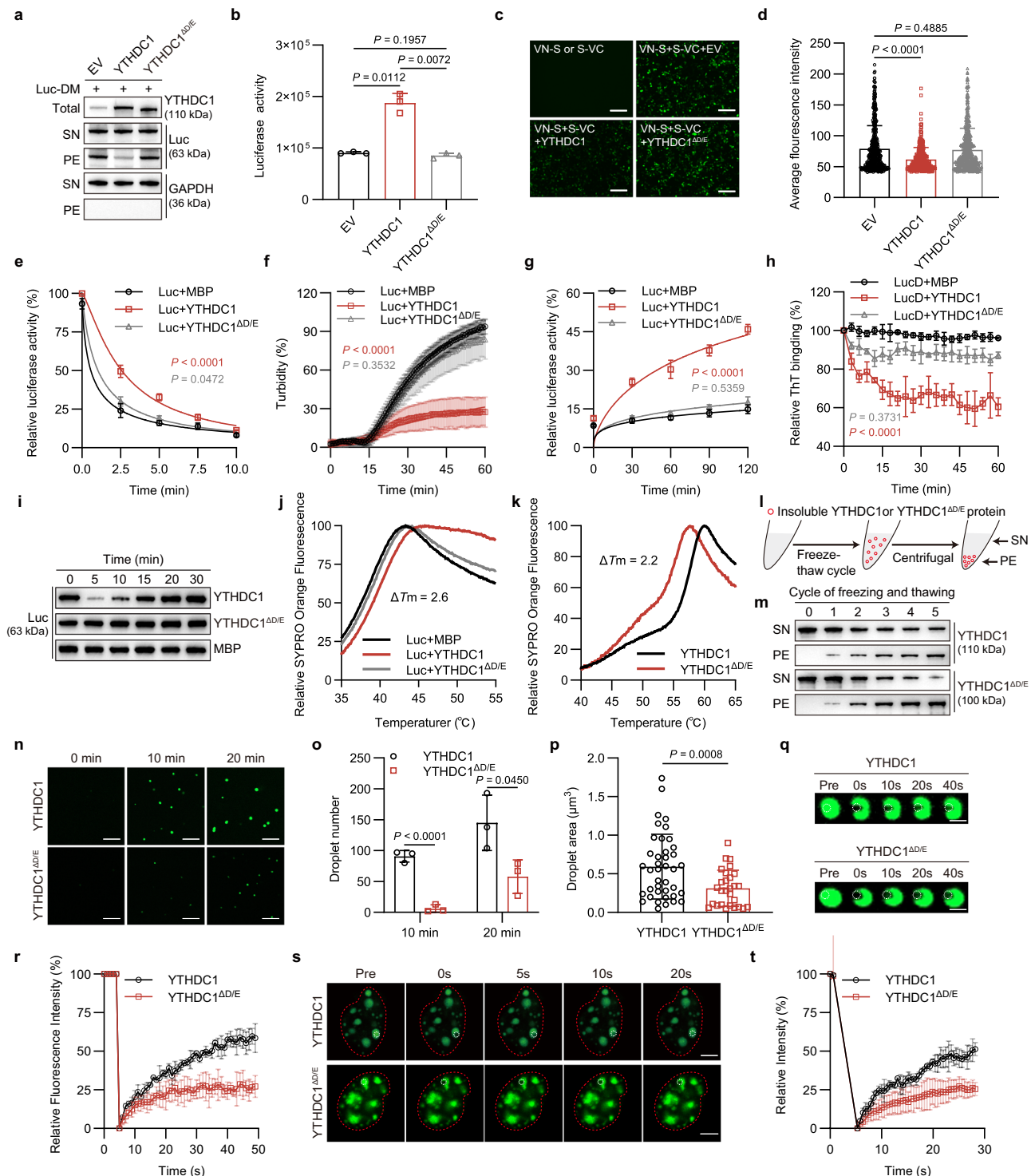
Fig. 1 | YTHDC1 prevents protein misfolding and dissolves protein aggregates.

a Heat-denatured luciferase was incubated with MBP, YTHDC1, or HSP70/40, and enzymatic activity was measured over time. **b** Turbidity of heat-denatured luciferase monitored by OD₆₀₀. **c** ThT fluorescence of monomeric misfolded LucD incubated with YTHDC1. **d** Schematic of LucD conformational states and trypsin sensitivity. **e** Trypsin digestion of LucD after incubation with the indicated proteins. **f, g** Sedimentation assays of luciferase aggregates incubated with YTHDC1 for the indicated times (**f**) or concentrations (**g**); supernatant (SN) and pellet (PE) fractions were analyzed by immunoblotting. **h** Reactivation of heat-denatured luciferase by the indicated proteins. **i** Michaelis–Menten analysis of luciferase reactivation kinetics. **j** Luciferase activity in HEK293T cells expressing Luc-DM together with empty vector (EV) or YTHDC1. **k** Sedimentation assay of Luc-DM in HEK293T cells expressing EV or YTHDC1. **l** PROTEOSTAT staining of control and

YTHDC1-depleted HeLa cells. Nuclei were counterstained with DAPI. Scale bar, 20 μ m. **m** Quantification of PROTEOSTAT fluorescence. **n** Bimolecular fluorescence complementation (BiFC) assay showing α -synuclein oligomerization in HEK293T cells expressing EV or YTHDC1. Scale bar, 100 μ m. **o** Quantification of BiFC fluorescence. **p** Domain organization and intrinsic disorder prediction of YTHDC1. **q** Schematic of wild-type YTHDC1 and mutants. **r** Sedimentation assay of Luc-DM in HEK293T cells expressing EV, YTHDC1, YTHDC1^W, or YTHDC1^{WW}. **s** Luciferase activity in HEK293T cells expressing the indicated constructs. Data are mean $\pm$ SD ($n = 3$ independent biological replicates) unless otherwise indicated. Statistical analyses: extra sum-of-squares F test (**a**), ANCOVA (**b, c**), two-tailed unpaired Student's t -test (**j, m, o**), and one-way ANOVA with Tukey's post hoc test (**s**). Experiments in (**e–g, k**, and **r**) were independently repeated three times with similar results. Source data are provided with this paper.

observed a lower T_m for YTHDC1^{AD/E} itself compared with wild-type (Fig. 2k) indicating compromised structural integrity and a reduced ability to withstand thermal denaturation. Freeze-thaw sedimentation further revealed reduced solubility of YTHDC1^{AD/E}, suggesting that YTHDC1 maintains superior stability compared with YTHDC1^{AD/E} (Fig. 2l, m), suggesting that polyD/E region involved in self-stability.

To further investigate the stability and biophysical properties of YTHDC1 influenced by polyD/E sequence, we examined the condensation and mobility of the proteins using purified recombinant YTHDC1-GFP and YTHDC1^{AD/E}-GFP in vitro (Supplementary Fig. 2j). With RNA present, YTHDC1-GFP underwent faster and more extensive phase separation than YTHDC1^{AD/E} (Fig. 2n, o), forming larger liquid droplets (Fig. 2p). Fluorescence recovery after photobleaching (FRAP)



analysis demonstrated impaired fluidity of YTHDC1^{ΔD/E}-GFP condensates compared to those formed by YTHDC1-GFP, indicating intrinsic differences in material properties between the two variants in vitro (Fig. 2q, r). Intriguingly, we also observed that compared with wild-type, YTHDC1^{ΔD/E} formed an amorphous shape of puncta in cellular environments with lower mobility (Fig. 2s, t). To exclude the possibility that the loss of chaperone function in YTHDC1^{ΔD/E} is caused by altered phase separation morphology, we examined two additional YTHDC1 mutants, YTHDC1^W and YTHDC1^{WW}, which were previously shown in cellular assays to retain chaperone activity comparable to that of wild-type YTHDC1 (Fig. 1p–s). YTHDC1^W and YTHDC1^{WW} were more diffused than YTHDC1^{ΔD/E} (Supplementary Fig. 2k), further

distinguishing phase separation morphology from chaperone function. Collectively, these results demonstrate that loss of the polyD/E tract compromises the stability and mobility of YTHDC1, thereby likely attenuating its chaperone capacity, and that this functional impairment is not directly attributable to alterations in phase separation morphology.

hnRNP A1 is a candidate client of YTHDC1

To identify physiological client proteins of YTHDC1 acting as a molecular chaperone in vivo, we conducted a comprehensive proteomic profiling by overexpressing Flag-YTHDC1 or Flag alone as a control in HEK293T cells, followed by immunoprecipitation–mass spectrometry

Fig. 2 | The polyD/E region mediates the chaperone function of YTHDC1.

a Sedimentation assay of Luc-DM in HEK293T cells expressing EV, YTHDC1, or YTHDC1^{ΔD/E}. **b** Luc-DM activity in HEK293T cells expressing the indicated constructs. **c** Representative BiFC images of α-synuclein oligomerization in HEK293T cells expressing EV, YTHDC1, or YTHDC1^{ΔD/E}. Scale bar, 100 μm. **d** Quantification of BiFC intensity (EV: *n* = 446, YTHDC1: *n* = 341, YTHDC1^{ΔD/E}: *n* = 480 cells). **e** Activity of heat-denatured luciferase incubated with MBP, YTHDC1, or YTHDC1^{ΔD/E}. **f** Turbidity of heat-denatured luciferase monitored by OD₆₀₀. **g** Reactivation of heat-denatured luciferase by the indicated proteins. **h** ThT fluorescence of monomeric misfolded LucD incubated with YTHDC1 or YTHDC1^{ΔD/E}. **i** Time-dependent trypsin sensitivity of LucD after incubation with the indicated proteins. **j** Thermal shift analysis of native luciferase incubated with YTHDC1 or YTHDC1^{ΔD/E} at the indicated molar ratios. **k** Thermal shift analysis of YTHDC1 and YTHDC1^{ΔD/E}. **l** Schematic of freeze-thaw treatment and sedimentation to separate

SN and PE fractions. **m** Sedimentation analysis of YTHDC1 and YTHDC1^{ΔD/E} following freeze-thaw treatment. **n** Time-lapse images of RNA-induced phase separation of YTHDC1 and YTHDC1^{ΔD/E}. Scale bar, 20 μm. **o**, **p** Quantification of droplet number (**o**) and size (**p**). **q**, **r** In vitro fluorescence recovery after photobleaching (FRAP) assays. Representative images (**q**) and quantification of fluorescence recovery (**r**). White dashed lines indicate bleached regions. Scale bar, 2 μm. **s**, **t** FRAP assays in HeLa cells. Representative images (**s**) and quantification of fluorescence recovery (**t**). Red dashed lines outline nuclei and white dashed lines indicate bleached regions. Scale bar, 10 μm. Data are mean ± SEM (**e**, **g**) or mean ± SD (**b**, **d**, **o**, **p**) from *n* = 3 independent experiments. Statistical analyses: one-way ANOVA with Tukey's post hoc test (**b**, **d**), two-tailed unpaired Student's *t*-test (**o**, **p**), extra sum-of-squares *F* test (**e**, **g**), and ANCOVA (**f**, **h**). Experiments in **a** and **m** were independently repeated three times with similar results. Source data are provided with this paper.

(IP-MS) analysis. Among all candidates, we focused on proteins reported to be prion-like and physiologically relevant while excluding those closely associated with YTHDC1's core RNA-metabolic axis¹⁰, nominating hnRNPA1, an RNA-binding protein involved in pre-mRNA processing and alternative splicing (Fig. 3a). In addition, in human neuroblastoma SH-SY5Y cells, IP-MS analysis following overexpression of Flag-hnRNPA1 consistently revealed a relatively high abundance of YTHDC1 protein (Fig. 3b). The pathological amyloid fibril formation by hnRNPA1 is strongly linked to familial ALS, with disease-associated mutants hnRNPA1^{D262N} and hnRNPA1^{D262V} exhibiting an increased propensity for self-aggregation^{14,25}. Co-IPs confirmed interactions between YTHDC1 and hnRNPA1 (WT, D262N, D262V) (Supplementary Fig. 3a–c). BiFC further revealed nuclear complementation between YTHDC1 and hnRNPA1 in HEK293T cells (Fig. 3c). Collectively, these findings indicate that hnRNPA1 likely represents a physiological client of the YTHDC1 chaperone machinery in cells.

YTHDC1 stabilizes hnRNPA1 condensates and suppresses fibrillation

Recent studies have indicated that hnRNPA1 can undergo liquid-liquid separation, followed by irreversible solidification, and ultimately fibril formation^{26,27}. This pathological fibrillation is thought to be a central driver of ALS¹². To investigate the role of YTHDC1 in this process, we purified recombinant hnRNPA1 proteins encompassing both wild-type and disease-associated mutants (hnRNPA1^{D262N} and hnRNPA1^{D262V}) (Supplementary Fig. 3d). Purified wild-type or mutant hnRNPA1 proteins were incubated with YTHDC1 and assessed for phase separation analysis. YTHDC1 maintained spherical, stable condensates, whereas MBP negative control showed progressive aggregation (Fig. 3d and Supplementary Fig. 3e). Fluorescence intensity distribution analysis further demonstrated that YTHDC1 colocalized with ALS-linked mutant, hnRNPA1^{D262N} and enriched in the core of droplets (Fig. 3e), reminiscent of HSP70 partitioning in TDP-43 assemblies, suggesting conserved proteostatic mechanisms²⁸. FRAP assays further showed partial recovery of condensates treated with YTHDC1 but negligible recovery in controls, indicating increased fluidity from the gel-like state (Fig. 3f–i and Supplementary Fig. 3f, g). ThT kinetics demonstrated that YTHDC1 delayed the fibril nucleation of hnRNPA1 disease mutants (Figs. 3j and S3h), these results were further corroborated by transmission electron microscopy (TEM) (Fig. 3k and Supplementary Fig. 3i). Collectively, these findings indicate that YTHDC1 effectively maintains the stability of hnRNPA1 condensates in vitro, prevents their irreversible solidification, and thereby inhibits pathological fibril formation.

We next sought to examine the effect of YTHDC1 on hnRNPA1 in vivo. We first knocked down YTHDC1 in HeLa cells using siRNA, lysed the cells with RIPA buffer, removed debris by centrifugation, and incubated the supernatant with ThT at room temperature. Overall, protein amyloid aggregates were assessed by ThT fluorescence intensity. YTHDC1 knockdown led to a significant increase in cellular

protein amyloid aggregates (Fig. 3l). Sedimentation assays further revealed a marked accumulation of hnRNPA1 in the NP-40-insoluble fraction, accompanied by a significant reduction in the soluble fraction (Fig. 3m). We then generated HeLa cell lines stably expressing wild-type or mutant hnRNPA1 under a doxycycline (DOX)-inducible promoter. Upon DOX induction, insoluble hnRNPA1 levels increased in a time-dependent manner for both wild-type and mutants, with the two disease-linked mutants displaying a particularly pronounced increase (Supplementary Fig. 4a). This accumulation of insoluble hnRNPA1 was paralleled by a significant rise in overall protein amyloid aggregates (Supplementary Fig. 4b). Overexpression of YTHDC1 in these stable lines markedly reduced insoluble hnRNPA1 levels and concomitantly lowered overall protein amyloid aggregates (Fig. 3n, o). In stark contrast, the YTHDC1^{ΔD/E} mutant failed to exhibit any detectable activity in this assay (Supplementary Fig. 4c, d).

Previous reports have established that hnRNPA1 can excessively partition into stress granules (SGs) and transition into an irreversible solid-like state, ultimately forming pathogenic fibrils¹⁴. This process closely mirrors the in vitro trajectory of hnRNPA1 from liquid-liquid phase separation to irreversible fibrillation^{26,27}. Notably, the extent of hnRNPA1 recruitment to SGs is tightly correlated with its intrinsic aggregation propensity, with disease-associated mutants showing greater SG incorporation than wild-type. Thus, aberrant pathological accumulation of hnRNPA1 in SGs represents a pivotal intermediate driving irreversible amyloid fibrillogenesis and poses significant neurotoxic risks. We therefore investigated the role of YTHDC1 in the SG recruitment of hnRNPA1 and its mutants. Upon sodium arsenate-induced cellular stress, both wild-type hnRNPA1 and the pathological mutants translocated to TIAR-positive SGs (Fig. 3p and Supplementary Fig. 4e). YTHDC1 overexpression significantly reduced SG recruitment of both wild-type and mutant hnRNPA1 (Fig. 3p, q and Supplementary Fig. 4e). Collectively, YTHDC1 preserves the condensate dynamics of hnRNPA1 both in vitro and in cells and restrains the aggregation and fibrillation of pathological hnRNPA1.

YTHDC1 mitigates impaired neurite growth in primary neurons

To investigate the neuropathological effects of YTHDC1 on hnRNPA1 and its disease-associated mutants (hnRNPA1^{D262N} and hnRNPA1^{D262V}), we established a disease model with mouse primary cortical neurons. Primary cortical neurons from timed-pregnant C57BL/6 mice at embryonic day 15 (E15) were transduced with adeno-associated virus vector 9 (AAV9) expressing hnRNPA1, hnRNPA1^{D262N}, hnRNPA1^{D262V} or GFP control (Fig. 4a). One-week post-infection, hnRNPA1^{D262N} and hnRNPA1^{D262V} induced pronounced neuronal atrophy in primary neurons, whereas wild-type hnRNPA1 exhibited minimal effects (Fig. 4b). Sholl analysis and simple neurite tracing (ImageJ) quantified reduced synaptic complexity and neurite length of mutants (Fig. 4c, d)^{29,30}, indicating a strong neurotoxic effects caused by hnRNPA1^{D262N} and hnRNPA1^{D262V} while wild-type hnRNPA1 showed no detectable impairment.

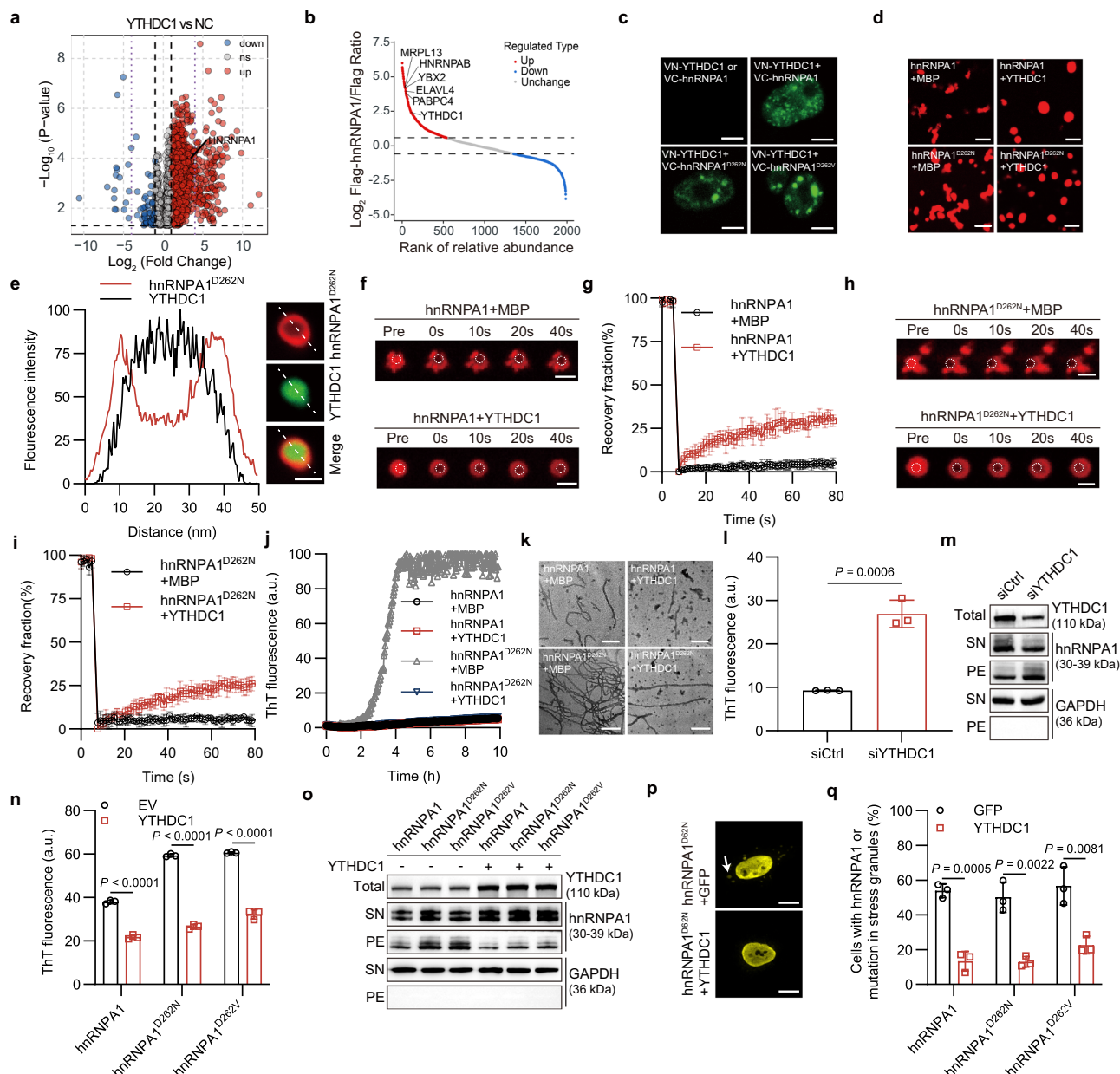
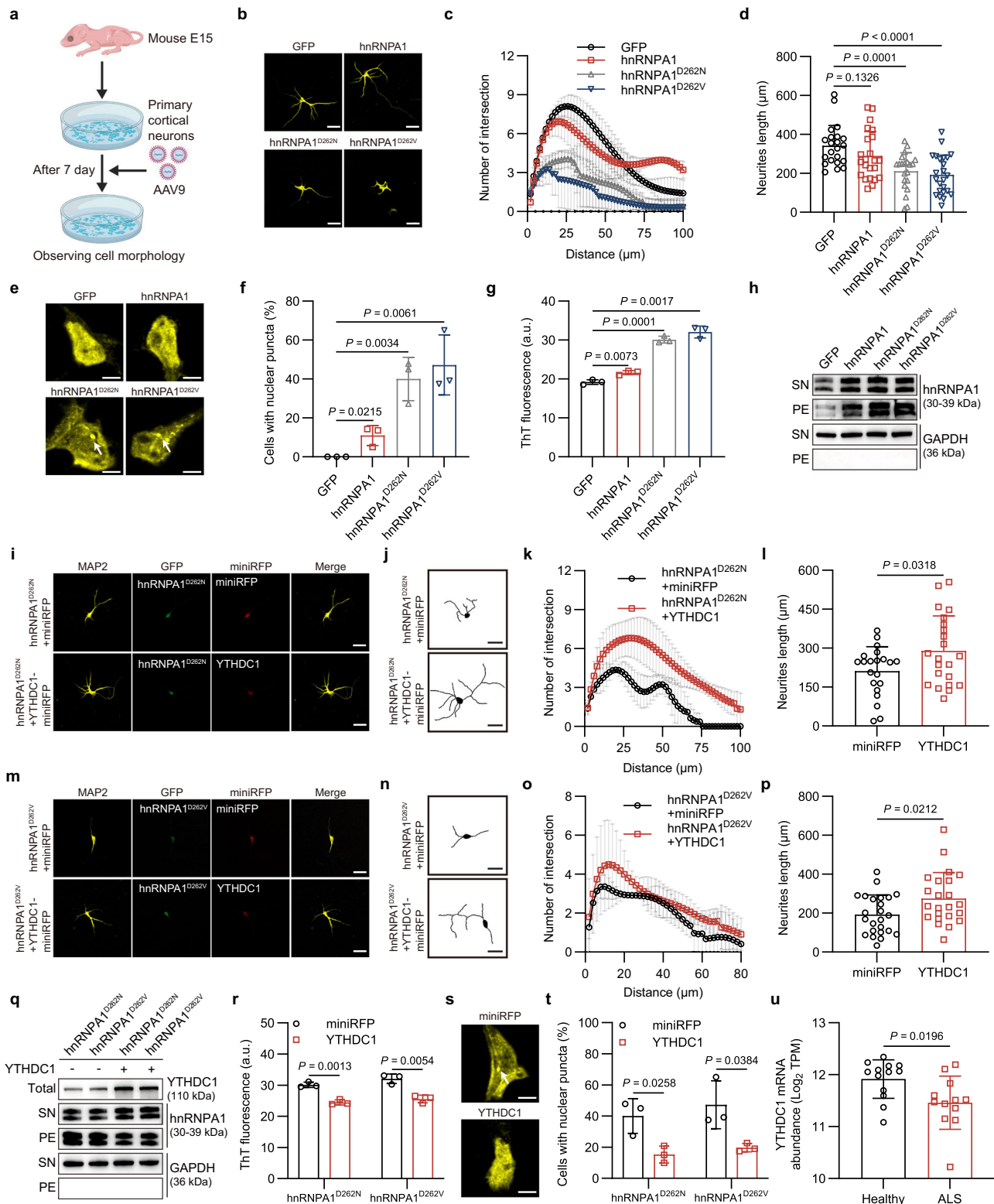


Fig. 3 | YTHDC1 suppresses pathological hnRNP A1 fibrillization and stress granule recruitment. **a** Volcano plot of proteins enriched by YTHDC1 immunoprecipitation (IP) versus control. Proteins with FC > 2 and $P < 0.05$ are highlighted. **b** Ranked abundance of nuclear proteins interacting with hnRNP A1 in SH-SY5Y cells; the top five proteins and YTHDC1 are shown, with Flag as a negative control. **c** BIFC analysis of interactions between YTHDC1 and wild-type (WT) or mutant hnRNP A1. Scale bar, 5 μ m. **d** Representative images of hnRNP A1 and hnRNP A1^{D262N} condensates after incubation with MBP or YTHDC1. Scale bar, 5 μ m. **e** Fluorescence intensity distribution and representative images of labeled YTHDC1 and hnRNP A1^{D262N} within condensates. Scale bar, 2 μ m. **f–i** FRAP assays of hnRNP A1 condensates. Representative images (**f**, **h**) and quantification of fluorescence recovery (**g**, **i**). White dashed lines indicate bleached regions. Scale bar, 2 μ m. **j** Thioflavin T (ThT) analysis of fibrillation of WT or mutant hnRNP A1 incubated with

YTHDC1. **k** Representative electron micrographs of hnRNP A1 and hnRNP A1^{D262N}. Scale bar, 100 nm. **l**, **m** Aggregation analyses in control or YTHDC1-depleted HeLa cells. ThT fluorescence (**l**) and sedimentation analysis of endogenous hnRNP A1 (**m**). **n**, **o** Aggregation analyses in HeLa cells stably expressing WT or mutant hnRNP A1 with or without YTHDC1 overexpression. ThT fluorescence (**n**) and sedimentation analysis of hnRNP A1 (**o**). a.u. arbitrary units. **p** Representative images showing reduced stress granule (SG) recruitment of hnRNP A1^{D262N} upon YTHDC1 expression. Scale bar, 10 μ m. **q** Quantification of cells displaying WT or mutant hnRNP A1 recruitment into SGs. Data are mean $\pm$ SD from $n = 3$ independent experiments. Two-tailed unpaired Student's t -test was used for (**l**, **n**, and **o**). Experiments in (**m** and **o**) were independently repeated three times with similar results. Source data are provided with this paper.

Concurrently, immunofluorescence imaging revealed that the two disease-associated hnRNP A1 mutants (hnRNP A1^{D262N} and hnRNP A1^{D262V}) formed nuclear puncta in a high proportion of primary neurons, whereas such structures were observed in only a small fraction of cells expressing wild-type hnRNP A1 and were entirely absent in GFP-expressing controls (Fig. 4e, f and Supplementary Fig. 5a). Using the same approach as described above, we lysed the transduced primary

neurons and measured overall protein amyloid aggregates by ThT fluorescence. Overexpression of hnRNP A1 (wild-type or mutants) led to a significant increase in total amyloid aggregates relative to the GFP control, with the two disease mutants inducing markedly higher amyloid aggregate levels (Fig. 4g). We further examined the levels of detergent-insoluble hnRNP A1. Consistent with the amyloid aggregation results, insoluble hnRNP A1 was dramatically elevated in neurons



expressing the mutants, while wild-type hnRNPA1 showed only modest accumulation (Fig. 4h and Supplementary Fig. 5b). These findings indicate that the two disease-linked hnRNPA1 mutants exert pronounced neurotoxicity in primary neurons, likely mediated by their heightened propensity for protein aggregation.

We subsequently manipulated the YTHDC1 levels in primary cortical neurons to systematically evaluate its capacity to reduce the hnRNPA1 mutant-induced synaptic pathology. We co-transduced primary cortical neurons with AAV9-YTHDC1 or AAV9-miniRFP in

conjunction with AAV9-hnRNPA1^{D262N} or AAV9-hnRNPA1^{D262V} at matched viral titers³¹. One week post-transduction, synaptic architecture was quantitatively evaluated that YTHDC1 co-expression significantly mitigated hnRNPA1^{D262N} driven synaptic impairments in primary cortical neurons compared to controls (Fig. 4i, j), with recovery of both synaptic complexity and neurites arborization length was also observed after YTHDC1 overexpression (Fig. 4k, l). Similarly, YTHDC1 exhibited a comparable protective effect against the impaired neurite growth induced by hnRNPA1^{D262V} in primary cortical neurons (Fig. 4m–p).

Fig. 4 | YTHDC1 mitigates impaired neurite growth caused by hnRNPA1 mutations. **a** Experimental design for AAV9 transduction of primary cortical neurons with GFP or WT or mutant hnRNPA1. Created with BioGDP.com. **b** Representative neurite morphology. Scale bar, 50 μ m. **c** Sholl analysis of neurite intersections. **d** Quantification of total neurite length ($n = 22, 23, 21$, and 24 cells for GFP, hnRNPA1, hnRNPA1^{D262N}, and hnRNPA1^{D262V}, respectively). **e** Representative images of nuclear hnRNPA1 puncta (arrowheads) in mutant-expressing neurons. Scale bar, 500 nm. **f** Quantification of cells with nuclear puncta ($n = 63, 63, 65$, and 67 , respectively). **g, h** Aggregation analyses in neurons expressing the indicated constructs. ThT fluorescence (**g**) and sedimentation analysis of hnRNPA1 (**h**). **i–l** Effects of YTHDC1 on neurite growth in hnRNPA1^{D262N}-expressing neurons. Representative morphology (**i**), neurite tracing (**j**), Sholl analysis (**k**), and total neurite length (**l**; miniRFP: $n = 21$, YTHDC1: $n = 22$). Scale bar, 50 μ m. **m–p** Corresponding analyses of hnRNPA1^{D262V}-expressing neurons. Representative morphology (**m**),

neurite tracing (**n**), Sholl analysis (**o**), and total neurite length (**p**; miniRFP: $n = 24$, YTHDC1: $n = 23$). Scale bar, 50 μ m. **q, r** Sedimentation analysis of hnRNPA1 (**q**) and ThT fluorescence (**r**) in neurons co-expressing hnRNPA1 mutants with miniRFP or YTHDC1. a.u. arbitrary units. **s** Representative nuclear hnRNPA1 puncta (arrowheads) in neurons expressing hnRNPA1 mutants with miniRFP or YTHDC1. Scale bar, 500 nm. **t** Quantification of cells with nuclear puncta ($n = 65, 67, 64$, and 66 cells, respectively). **u** Reanalysis of RNA-seq data (GSE234991) showing YTHDC1 transcript levels ($\log_2$ TPM) in dermal fibroblasts from healthy donors ($n = 12$) and ALS patients ($n = 11$). Data are mean $\pm$ SD from three independent experiments. One-way ANOVA with Tukey's post hoc test was used for (**d, f**, and **g**); two-tailed unpaired Student's *t*-test for (**i, p, r, t**, and **u**). Experiments in (**h** and **q**) were independently repeated three times with similar results. Source data are provided with this paper.

We further examined the levels of insoluble hnRNPA1 in these neurons. YTHDC1 co-expression led to a marked reduction in NP-40-insoluble hnRNPA1 and a corresponding increase in the soluble fraction relative to controls (Fig. 4q and Supplementary Fig. 5c). Concurrently, ThT fluorescence assays revealed a decrease in overall protein amyloid aggregates (Fig. 4r). Moreover, the proportion of primary neurons harboring hnRNPA1 mutant-induced nuclear puncta was significantly reduced upon YTHDC1 expression, with puncta disappearing in the majority of cells (Fig. 4s, t and Supplementary Fig. 5d). In contrast, overexpression of the YTHDC1^{AD/E} mutant failed to rescue impaired neurite growth triggered by the disease-linked hnRNPA1 mutants, restore hnRNPA1 solubility, or reduce total protein amyloid aggregates (Supplementary Fig. 6). These results suggest that YTHDC1 alleviates neurotoxicity caused by hnRNPA1 aggregation by promoting the resolubilization and functional recovery of insoluble hnRNPA1, and they further underscore the essential role of the polyD/E sequence in YTHDC1's chaperone function. Collectively, these data establish a molecular chaperone-like activity of YTHDC1 in counteracting hnRNPA1 mutation-associated synaptic degeneration, highlighting its neuroprotective role in proteinopathy-associated neurodegeneration.

Furthermore, analysis of publicly available transcriptomic data from ALS patient samples revealed a significant decrease in YTHDC1 expression (Fig. 4u)³². Given that hnRNPA1 mutations may be related to ALS pathogenesis, this reduction, together with our finding that YTHDC1 exerts a chaperone-mediated protective effect on hnRNPA1, reinforces the importance of YTHDC1 in the initiation and progression of ALS.

CK2-mediated phosphorylated impairs YTHDC1 chaperone activity

Based on the aforementioned research findings, we systematically investigated the regulatory mechanisms underlying the chaperone activity of YTHDC1 associated with the polyD/E region. IP-MS analysis comparing wild-type YTHDC1 and YTHDC1^{AD/E} revealed 89 proteins preferentially interacting of only with wild-type YTHDC1 ($P < 0.01$; $\log_2$ fold-change > 1.5), including all three subunits of protein kinase CK2 complex, Casein kinase II subunit α (CK2 α), Casein kinase II subunit α' (CK2 α') and Casein kinase II subunit β (CK2 β) (Fig. 5a and Supplementary Fig. 7a). The protein kinase CK2 complex is a constitutively active serine/threonine-protein kinase complex, and has been implicated in the aggregation of pathological proteins in several neurodegenerative diseases^{33–35}. CK2 is structurally composed of a tetrameric enzyme, consisting of two catalytic subunits, either homodimers (CK2 α or CK2 α') or heterodimers (CK2 α with CK2 α') and two non-catalytic β subunits (CK2 β)³⁶. Co-immunoprecipitation (co-IP) confirmed diminished binding to all three subunits of CK2 complex, including CK2 α , CK2 α' and CK2 β , compared to the wildtype YTHDC1 (Supplementary Fig. 7b–d). In vitro thiophosphate ester kinase assay utilizing cellularly precipitated CK2 α , which serves as the enzymatic component of the CK2 kinase complex, demonstrated that CK2 α

directly phosphorylates YTHDC1 (Fig. 5b), establishing YTHDC1 as a CK2 substrate via polyD/E-dependent engagement.

Next, we investigated whether phosphorylation by CK2 influences YTHDC1's chaperone or m⁶A reader function. Pharmacological inhibition of YTHDC1 with TBB, a highly selective ATP/GTP-competitive inhibitor of CK2³⁷, enhanced YTHDC1-mediated suppression of Luc-DM aggregation in the pellet and increased luciferase activity in a dose-dependent manner (Fig. 5c and Supplementary Fig. 7e). CK2 α knock-down by siRNA recapitulated these effects (Fig. 5d and Supplementary Fig. 7f–i). These results indicate that CK2-mediated phosphorylation constrains YTHDC1 chaperoning. RIP-seq following CK2 α depletion by siCK2 α showed over 90% overlap in YTHDC1-bound transcripts versus the control group, with minimal differences in binding to m⁶A-annotated mRNAs^{38–40} (Fig. 5e, f and Supplementary Fig. 8a) and no significant changes in canonical YTHDC1 downstream targets²⁴ (Fig. 5g). Thus, CK2 does not materially alter YTHDC1's m⁶A reader activity.

Phosphoproteomics and motif analyses by GPS 6.0 identified S146, T148, and S152 as candidate CK2 sites adjacent to the polyD/E proximal region^{41,42} (Supplementary Fig. 8b, c). Meanwhile, the evolutionary conservation of these amino acid residues in YTHDC1 across diverse species underscores their potential functional significance (Supplementary Fig. 8d, e). To elucidate the functional relevance of these sites, we generated both phosphorylation-mimetic (YTHDC1^{P-M}) and phosphorylation-deficient (YTHDC1^{P-D}) mutants of YTHDC1 (Fig. 5h). In vitro, YTHDC1^{P-D} exhibited reduced phosphorylation (Fig. 5i and Supplementary Fig. 8f), and in cells, YTHDC1^{P-D} retained chaperone activity as shown by reduced Luc-DM aggregation and enhanced enzymatic activity, whereas YTHDC1^{P-M} lost this capability (Fig. 5j, k).

We next examined the effect of CK2 on YTHDC1 function in a cellular context. In HeLa cells stably expressing hnRNPA1^{D262N}, co-expression of YTHDC1 with CK2 α led to a marked increase in NP-40 insoluble hnRNPA1 levels relative to the control, accompanied by a corresponding elevation in total protein amyloid aggregates (Fig. 5l, m). Moreover, CK2 α significantly enhanced the recruitment of hnRNPA1 to SGs, with no concomitant change in total hnRNPA1 protein levels (Fig. 5n, o and Supplementary Fig. 8g). Taken together, our results indicate that CK2-mediated phosphorylation of YTHDC1 at S146, T148, and S152 inhibits its chaperone function.

Discussion

Our findings establish YTHDC1 as a previously unrecognized molecular chaperone that safeguards hnRNPA1 from aberrant phase transition and irreversible aggregation. In vitro, YTHDC1 preserves the fluidity of hnRNPA1 condensates and prevents their maturation into solid-like assemblies that seed pathological fibrils (Figs. 3 and 6)⁴³. In cells, YTHDC1 suppresses the accumulation of insoluble hnRNPA1 species and limits excessive recruitment of hnRNPA1 into SGs, thereby intercepting a critical route toward aggregation driven by condensate

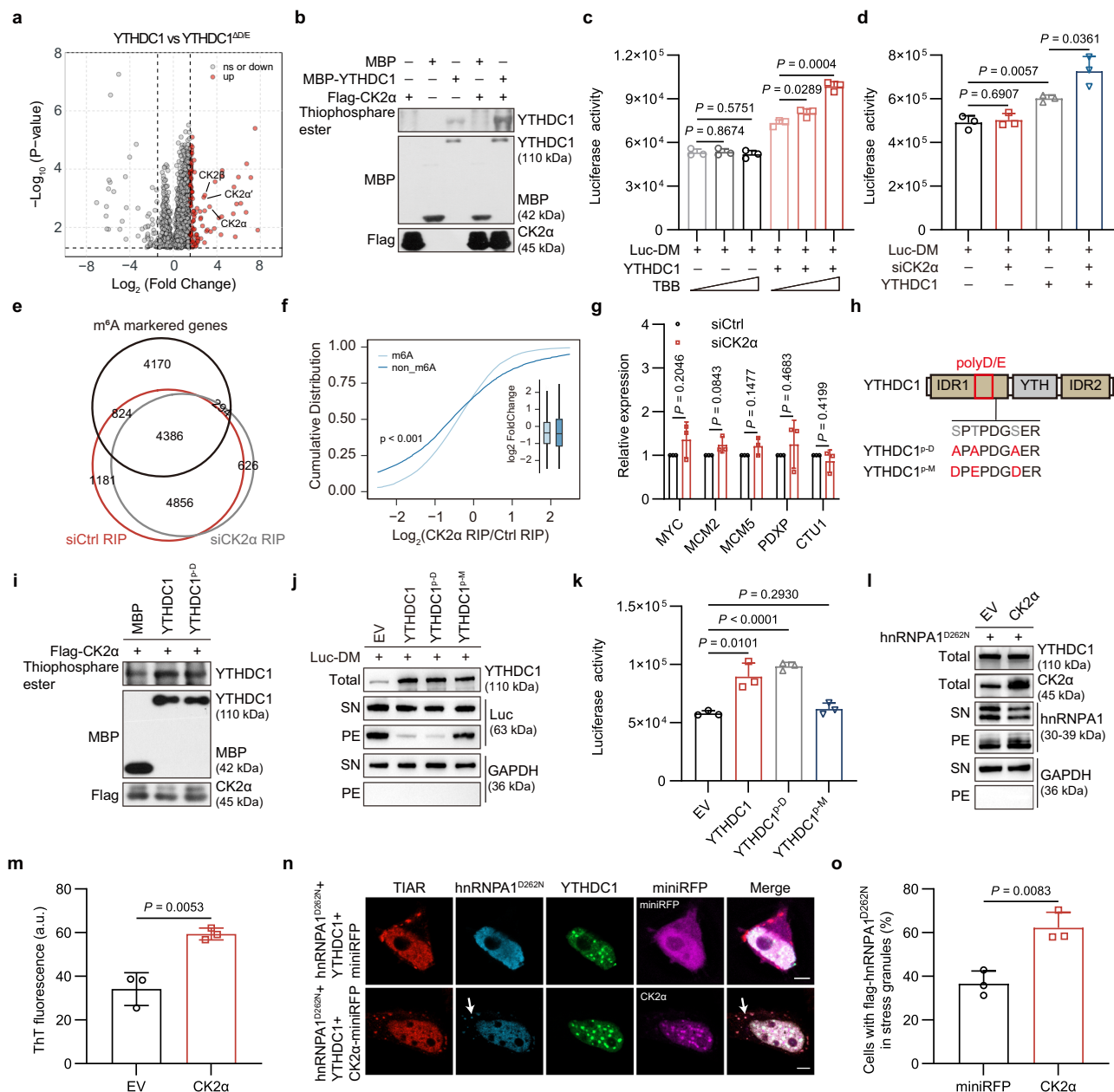


Fig. 5 | CK2-mediated phosphorylation impairs YTHDC1 chaperone activity.

a Volcano plot of proteins enriched in YTHDC1 versus YTHDC1^{ΔDE} immunoprecipitates. Dashed lines indicate $P < 0.01$ and $\log_2(FC) > 1.5$. **b** In vitro thiophosphate ester kinase assay showing CK2α-mediated phosphorylation of YTHDC1. **c** Luc-DM activity in YTHDC1-expressing HEK293T cells treated with the CK2 inhibitor TBB. **d** Luc-DM activity in control or CK2α-depleted HEK293T cells expressing EV or YTHDC1. **e** Overlap among YTHDC1 RIP targets, YTHDC1 RIP targets following CK2α depletion, and m⁶A-containing transcripts. (GSE29714, GSE55572, GSE87516, GSE90914, respectively). **f** Cumulative distribution and boxplots of m⁶A abundance in m⁶A ($n = 5504$) and non-m⁶A ($n = 6663$) transcripts. Whiskers indicate minimum to maximum, boxes indicate Q1–Q3, and center lines indicate medians. Two-sided Kolmogorov–Smirnov test, $D = 0.133$, $P = 1.64 \times 10^{-46}$; median difference 0.172 (bootstrap 95% CI, 0.118–0.219). **g** qPCR analysis of YTHDC1 target transcripts following CK2α depletion. **h** Domain organization of YTHDC1 and phosphorylation-

site mutants. p-M phospho-mimetic, p-D phospho-dead. **i** In vitro kinase assay comparing CK2α-mediated phosphorylation of YTHDC1 and YTHDC1^{P-D}. **j** Sedimentation analysis of Luc-DM in HEK293T cells expressing the indicated YTHDC1 constructs. **k** Luc-DM activity in HEK293T cells expressing the indicated constructs. **l**, **m** Aggregation analyses in HeLa cells stably expressing WT or mutant hnRNP A1 with CK2α. Sedimentation analysis (**l**) and ThT fluorescence (**m**). **n** Representative images of SG recruitment of hnRNP A1^{D262N} in HeLa cells expressing YTHDC1 with miniRFP or CK2α-miniRFP. Scale bar, 5 μm. **o** Quantification of cells with hnRNP A1^{D262N}-positive SGs. Data are mean $\pm$ SD from $n = 3$ independent experiments. Statistical analyses: one-way ANOVA with Tukey's post hoc test (**c**, **d**, **k**) and two-tailed unpaired Student's *t*-test (**g**, **m**, **o**). Experiments in (**b**, **i**, **j**, and **l**) were independently repeated three times with similar results. Source data are provided with this paper.

solidification (Figs. 3 and 6). In primary cortical neurons, YTHDC1 similarly reduces insoluble hnRNP A1 accumulation and nuclear puncta while rescuing neurite growth defects induced by disease-associated hnRNP A1 mutants (Figs. 4 and 6). When YTHDC1 was depleted, insoluble endogenous hnRNP A1 levels increased markedly, accompanied

by a significant rise in the global cellular aggregate burden, supporting a role for YTHDC1 in maintaining hnRNP A1 proteostasis at endogenous expression levels (Figs. 11, m and 31, m). Notably, disease-associated mutations in the PrLD of hnRNP A1 have been identified in rare familial cases of multisystem proteinopathy and ALS^{14,25}. However, as hnRNP A1

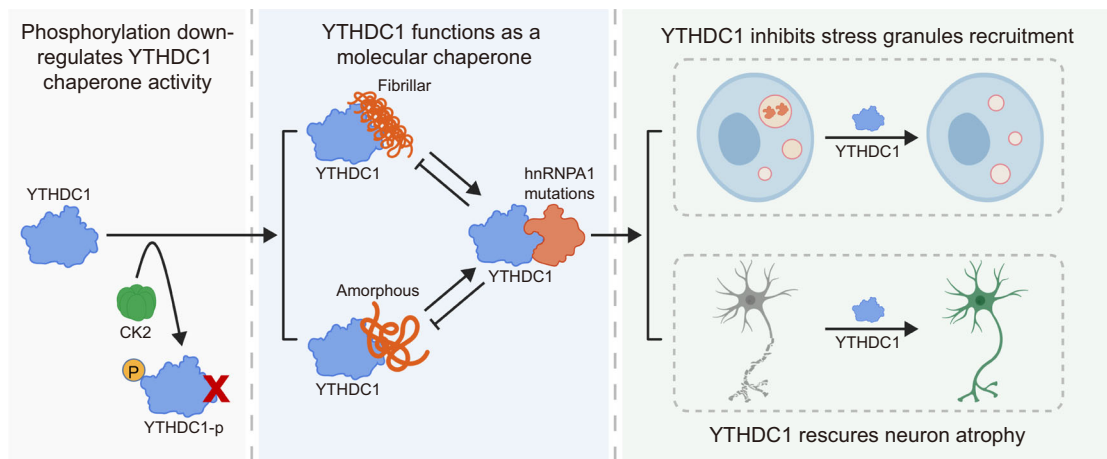


Fig. 6 | Schematic model of YTHDC1-mediated protein quality control. YTHDC1 functions as an ATP-independent molecular chaperone that suppresses the formation of amorphous and fibrillar hnRNP A1 aggregates and resolubilizes pre-formed protein aggregates. CK2-mediated phosphorylation negatively regulates

YTHDC1 chaperone activity. YTHDC1 also limits excessive hnRNP A1 recruitment into stress granules and mitigates mutant hnRNP A1-induced impairment of neurite growth. Created with BioGDP.com.

mutation-associated ALS represents an exceptionally rare genetic subtype, publicly available patient datasets directly linking YTHDC1 to hnRNP A1 pathology remain limited. Analysis of a publicly available transcriptomic dataset³² from ALS patient samples nevertheless revealed significantly downregulated YTHDC1 expression (Fig. 4u). Although this observation does not establish a direct relationship between YTHDC1 and hnRNP A1 pathology in patients, combined with our evidence that YTHDC1 suppresses the accumulation of insoluble hnRNP A1, these data support a model in which reduced YTHDC1 expression may contribute to proteostasis impairment in specific ALS patient subgroups. Furthermore, CK2-mediated phosphorylation of YTHDC1 at polyD/E-proximal residues selectively suppresses its chaperone activity without affecting its canonical m⁶A-reader function, as phosphorylated YTHDC1 retains normal association with m⁶A-modified transcripts and downstream pathway regulation (Figs. 5 and 6). Together, these findings demonstrate that the proteostatic and RNA-metabolic functions of YTHDC1 are mechanistically separable.

Several features distinguish YTHDC1 from classical ATP-dependent chaperones such as the HSP family⁴⁴. First, YTHDC1 displays unusually restricted evolutionary conservation, with definitive orthologs identified primarily in higher eukaryotes, including humans, mice, rabbits, and *Drosophila*, whereas canonical HSP chaperones are broadly conserved across prokaryotes and eukaryotes (Supplementary Fig. 8d)⁴⁴. This restricted phylogenetic distribution suggests that the chaperone activity of YTHDC1 may have evolved relatively late to fulfill specialized proteostatic demands in complex cellular environments. Second, YTHDC1 exhibits marked client selectivity. Unlike broad-spectrum chaperones, YTHDC1 failed to suppress aggregation of several other neurodegeneration-associated proteins, including FUS(P525L), TDP-43, and ATXN2(poly85Q), indicating that its activity is not globally promiscuous (Fig. 3a and Supplementary Fig. 8h, i). Third, both YTHDC1 and hnRNP A1 are nuclear RNA-binding proteins occupying overlapping subcellular environments, enabling spatially restricted proteostasis surveillance. Finally, unlike ATP-dependent chaperones, YTHDC1 functions independently of ATP hydrolysis, thereby avoiding both long-range transport requirements and additional metabolic expenditure to maintain hnRNP A1 solubility (Fig. 1). Collectively, these features define a local, client-selective, and energetically efficient mode of chaperone action.

At first glance, it may seem unexpected that an m⁶A reader embedded within the RNA-processing machinery would evolve a dedicated chaperone function. However, increasing evidence suggests

that the functional boundary between RNA metabolism and PQC is more interconnected than previously appreciated. Kap β 2 (karyopherin- β 2), a nuclear transport receptor, has been shown to function as a potent chaperone and disaggregase for multiple aggregation-prone RNA-binding proteins, including hnRNP A1, by directly suppressing aberrant phase transition and fibrillization⁴⁵. Similarly, FUS, another ALS-associated RNA-binding protein, was recently reported to possess chaperone-like activity that maintains TAZ condensate dynamics and transcriptional activation⁴⁶. Together with our findings, these studies support an emerging model in which RNA-binding proteins and RNA-associated factors can moonlight as proteostatic regulators, thereby blurring the classical distinction between RNA metabolism and PQC.

An emerging theme highlighted by Kap β 2, FUS, and now YTHDC1 is that proteins classically assigned to RNA transport or RNA-processing pathways can simultaneously function as regulators of proteostasis. This raises an important conceptual question: how are the canonical and non-canonical chaperone functions coordinated within the same protein without creating functional interference? For Kap β 2, coordination appears to arise from direct coupling between nuclear transport and proteostasis surveillance. The canonical transport receptor function and the non-canonical chaperone activity are mechanistically intertwined, as substrate recognition for transport inherently suppresses pathological phase transition and fibrillization⁴⁵. By contrast, the recently described chaperone-like activity of FUS appears to rely more directly on its intrinsic condensation properties, the same low-complexity and IDRs that enable physiological condensate assembly may also provide local buffering capacity against aberrant solidification⁴⁶. Our findings suggest that YTHDC1 adopts yet another organizational strategy. We initially considered the possibility that CK2-dependent phosphorylation might function as a molecular switch balancing these two activities, as CK2 phosphorylates YTHDC1 at S146, T148, and S152 and selectively inhibits its chaperone activity (Fig. 5 and Supplementary Fig. 7)⁴⁷. However, disruption of CK2 activity did not measurably alter YTHDC1-dependent m⁶A pathway regulation, indicating that these two functions are regulated independently rather than through a simple binary switch (Fig. 5e–g and Supplementary Fig. 8a, b). Thus, the coordination of dual functionality likely relies on multiple regulatory mechanisms that remain to be fully elucidated^{48,49}. These examples suggest that multifunctional RNA-associated proteins may employ diverse strategies to integrate canonical cellular functions with local proteostasis regulation. This diversity highlights the evolutionary plasticity of cellular proteostasis systems and suggests that

local PQC may be deeply embedded within other multifunctional protein networks.

Collectively, our findings identify YTHDC1 as an unconventional, ATP-independent molecular chaperone specialized for maintaining the proteostatic integrity of aggregation-prone RNA-binding proteins. Rather than replacing the general PQC machinery, YTHDC1 appears to provide a spatially restricted surveillance mechanism tailored to vulnerable nuclear clients such as hnRNPA1. The convergence within a single protein of an m⁶A-reader module and a phosphorylation-regulated chaperone domain highlights an unexpected evolutionary strategy for coupling RNA metabolism with protein homeostasis. More broadly, our work expands the functional repertoire of RNA-binding proteins and reveals an additional layer of proteostasis regulation embedded within the epitranscriptomic machinery.

Methods

Ethical statement

All research procedures comply with all relevant ethical regulations. The study protocol was approved by the Institutional Animal Care and Use Committee (IACUC) of Nanjing University (Approval No. IACUC-2603014).

Plasmids

All plasmids were constructed using the pRK5 backbone unless otherwise specified. The following constructs were generated: YTHDC1-His, HSP104^{A503S}-His, YTHDC1^{ΔD/E}-His, YTHDC1^W-His, YTHDC1^{WW}-His, YTHDC1^D-His, YTHDC1^M-His, YTHDC1^{S146E}-His, YTHDC1^{T148D}-His, YTHDC1^{S152E}-His, Flag-YTHDC1, Flag-YTHDC1^{ΔD/E}, HA-CSNK2A1, HA-hnRNPA1, HA-hnRNPA1^{D262N}, HA-hnRNPA1^{D262V}, MBP-Tev-YTHDC1-His, MBP-Tev-YTHDC1^{ΔD/E}-His, MBP-Tev-YTHDC1-GFP-His, MBP-Tev-YTHDC1^{ΔD/E}-GFP-His, MBP-Tev-YTHDC1^D-His, MBP-Tev-hnRNPA1-His, MBP-Tev-hnRNPA1^{D262N}-His, and MBP-Tev-hnRNPA1^{D262V}-His. Plasmids encoding MBP and HSP104^{A503S} were generated in the pET28a vector. The plasmid pT7lucC-His expresses a C-terminally truncated variant of Photinus pyralis luciferase (LucDHIS6), in which the last 62 residues were replaced by the sequence SKLSYEQDGL-HAGSPAALF followed by a hexahistidine (6×His) tag. Plasmids pRK5-Luc-DM-His, VNS, and SVC have been described previously⁴. Constructs for the expression of GFP, YTHDC1-GFP, and YTHDC1^{ΔD/E}-GFP in mammalian systems were also as previously reported²⁴.

Antibodies

The following primary antibodies were used in this study: anti-Flag (MBL, Cat# M185-3, Mouse monoclonal), anti-His (Santa Cruz Biotechnology, Cat# sc-8036, Mouse monoclonal), anti-MBP (Proteintech, Cat# 66003-1-Ig, Mouse monoclonal), anti-HA (Proteintech, Cat# 51064-2-AP, Rabbit polyclonal), anti-α-synuclein (Proteintech, Cat# 10842-1-AP, Rabbit polyclonal), anti-hnRNPA1 (Proteintech, Cat# 11176-1-AP, Rabbit polyclonal), anti-FUS (Proteintech, Cat# 11570-1-AP, Rabbit polyclonal), anti-TDP-43 (Proteintech, Cat# 10782-2-AP, Rabbit polyclonal), anti-Beta Tubulin (Proteintech, Cat# 10094-1-AP, Rabbit polyclonal), anti-YTHDC1 (Abcam, Cat# ab122340, Rabbit polyclonal), anti-Firefly Luciferase (Abcam, Cat# ab21176, Rabbit polyclonal), anti-GFP (Proteintech, Cat# 50430-2-AP, Rabbit polyclonal), anti-GAPDH (Proteintech, Cat# 10494-1-AP, Rabbit polyclonal), anti-Lamin B1 (HUA BIO, Cat# ET1606-27, Rabbit monoclonal), anti-MAP2 (Abcam, Cat# ab183830, Rabbit monoclonal), anti-TIAR (Cell Signaling Technology, Cat# 8509S, Rabbit Monoclonal), and the oligomer-specific A11 antibody (Invitrogen, Cat# AHB0052, Rabbit polyclonal). For immunodetection, horseradish peroxidase (HRP)-conjugated anti-rabbit (Cell Signaling Technology, Cat# 7074S) and anti-mouse (Cell Signaling Technology, Cat# 7076S) secondary antibodies were used. For immunofluorescence, Goat anti-Rabbit IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor 594-conjugated (Invitrogen, Cat# A-11012) and Goat anti-Rabbit IgG (H+L) Highly Cross-Adsorbed

Secondary Antibody, Alexa Fluor 647-conjugated (Invitrogen, Cat# A-21245) secondary antibodies were employed. All antibodies were diluted at 1:2000 for western blotting. All antibodies were diluted at 1:200 for immunofluorescence.

Cell culture

HEK293T and HeLa cell lines were acquired from the American Type Culture Collection. Cell lines were authenticated by the supplier. These cells were cultured in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum (Gibco, Cat# A5669701) and 1% penicillin/streptomycin and maintained at 37 °C in a humidified incubator with 5% CO₂, unless stated otherwise. HEK293F cells were kindly provided by Dr. Xuepeng Wei and were cultured in SMM 293-TII Expression Medium (Sino Biological, Cat# M293TII) at 37 °C using a shaking incubator supplied with 8% CO₂.

Protein expression and purification

Recombinant HSP104^{A503S}-His₆ protein was expressed in *Escherichia coli* BL21(DE3) cells grown in lysogeny broth medium. Protein expression was induced with 0.2 mM isopropyl β-D-1-thiogalactopyranoside at an optical density at 600 nm (OD₆₀₀) of 0.6–0.8, followed by incubation at 18 °C for 24 h. Cells were harvested and lysed in Buffer A (50 mM Tris-HCl, pH 7.4, 1 M NaCl, 5% glycerol, 1 mM dithiothreitol (DTT), 1 mM phenylmethylsulfonyl fluoride (PMSF), and protease inhibitors (Roche, Cat# 4693132001)). The lysate was centrifuged at 15,000 × g for 20 min at 4 °C. The resulting supernatant was applied to a 1-mL HisTrap FF column (Cytiva, Cat# 17531901) on an ÄKTA Purifier system (GE Healthcare, USA). Proteins were eluted using a linear gradient from Buffer A to Buffer B (Buffer A supplemented with 500 mM imidazole). Eluted fractions were analyzed by sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE), and the molecular weight of the target protein was confirmed using a protein standard (Sparkjade, Cat# EC1019-A).

Constructs encoding MBP-Tev-YTHDC1-His, MBP-Tev-YTHDC1^{ΔD/E}-His, MBP-Tev-YTHDC1^D-His, MBP-Tev-hnRNPA1-His, MBP-Tev-hnRNPA1^{D262N}-His, and MBP-Tev-hnRNPA1^{D262V}-His were transfected into HEK293F cells. The cells were lysed in lysis buffer (50 mM Tris-HCl, pH 8.0, 500 mM NaCl, 5% glycerol, 1 mM DTT, 1 mM PMSF, 100 μg/mL RNase A, and protease inhibitors (Roche, Cat# 4693132001)). After centrifugation, the supernatants were incubated with Ni-NTA resin for 3 h at either 4 or 25 °C. The bound proteins were eluted using elution buffer (50 mM Tris-HCl, pH 8.0, 250 mM NaCl, 5% glycerol, 500 mM imidazole, and 1 mM DTT) and subsequently incubated with Dextrin Beads (Smart-Lifesciences, Cat# SA077100) overnight at 4 or 25 °C. Proteins were eluted from the beads using maltose-containing buffer (50 mM Tris-HCl, pH 8.0, 250 mM NaCl, 5% glycerol, 10 mM maltose, and 1 mM DTT), concentrated using storage buffer (20 mM HEPES, pH 7.3, 500 mM NaCl, 1 mM DTT), flash-frozen in liquid nitrogen, and stored at −80 °C.

Luciferase inactivation assay

10 or 50 nM luciferase (Sigma, Cat# L9420-1MG) was heated at 42 °C either alone or in the presence of the indicated proteins in luciferase refolding buffer (25 mM HEPES-KOH, pH 7.4, 150 mM potassium acetate, 10 mM magnesium acetate, and 10 mM DTT). HSP70-HSP40 was used as positive control, while the MBP protein served as a negative control. ATP-Mg²⁺ and an ATP-regeneration system (Enzo Life Sciences, Cat# BML-EW9810) were included in samples with HSP70-40 proteins, but not in samples with YTHDC1. HSP70 and HSP40 proteins were kindly gifted by Dr. Yajun Jiang⁵⁰. Luciferase activity was measured using the Luciferase Assay System (Promega, Cat# E1500) on a microplate reader (BioTek, USA). Data was collected using BioTek Gen 5 software and expressed as percentages relative to the native luciferase control.

Luciferase aggregation assay

600 nM luciferase (Sigma, Cat# L9420-1MG) was heated at 42 °C either alone or in the presence of the indicated proteins in HEPES buffer (50 mM HEPES-KOH, pH 7.4, 50 mM KCl, 5 mM MgCl₂, and 2 mM DTT). Luciferase aggregation was monitored by measuring the absorbance at 600 nm using a microplate reader (BioTek, USA).

Luciferase reactivation assay

Firefly luciferase (Sigma, Cat# L9420-1MG) was heat-inactivated at 42 °C for 10 min and then added to reactions at a final concentration of 50 nM in luciferase refolding buffer. Denatured luciferase was incubated with the indicated proteins at 25 °C for 90 min or for the specified durations. The luciferase activity of the reaction mixtures was analyzed using the Luciferase Assay System (Promega, Cat# E1500) on a microplate reader (BioTek, USA). The solubility of the reaction mixtures was assessed by western blot analysis following centrifugal separation into supernatant (SN) and pellet (PE). Steady-state kinetic analysis was performed by incubating 100 nM YTHDC1 or HSP104^{AS03S} with increasing concentrations of denatured luciferase monomers for 30 min. Luciferase activity was measured, and kinetic curves were fitted to determine $V_{\max}$ and K_m using nonlinear regression based on the Michaelis–Menten equation in GraphPad Prism 10.

Unfoldase assays for LucD

LucD was inactivated through repeated freeze-thaw cycles, and monomers were isolated as previously described in ref. 19. For the ThT assay, misfolded LucD monomers (3 μM) were incubated with varying concentrations of YTHDC1 and 20 μM ThT. ThT binding was quantified using a microplate reader (BioTek) with excitation and emission wavelengths set at 450 and 485 nm, respectively.

Trypsin digestion assay

Recombinant LucD (50 nM) was incubated either alone or in the presence of the specified proteins at 25 °C for varying time intervals. Samples were then treated with 2.5 μM trypsin at 22 °C for 3 min and analyzed by Western blotting.

Sedimentation assay for luciferase aggregation in cells

To investigate the effect of YTHDC1 on luciferase aggregation in cells, HEK293T cells were seeded in 6-well plates and cultured for 12 h. Cells were then transfected with Luc-DM and YTHDC1 using poly-ethylenimine (YEASEN, Cat# 40816ES02). After 24 h, cells were harvested and lysed in 200 μL of ice-cold lysis buffer 50 mM Tris, pH 8.8, 100 mM NaCl, 5 mM MgCl₂, 0.5% NP-40, 1 mM DTT, 250 IU/mL benzonase, 1 mM PMSF, and 1× complete protease inhibitor cocktail for 30 minutes on ice. Lysates were centrifuged at 15,000 × *g* for 30 min at 4 °C. The NP-40-soluble SN was collected as the soluble fraction. The NP-40-insoluble pellet was resuspended in 200 μL of ice-cold pellet buffer (20 mM Tris, pH 8.0, 15 mM MgCl₂, 1 mM DTT, 250 IU/mL benzonase, 1 mM PMSF, and 1× complete protease inhibitor cocktail) for 30 min on ice, followed by the addition of 50 μL of 5× loading buffer. Samples were heated at 95 °C for 10 minutes. The SDS-solubilized proteins from the PE were designated as the pellet fraction. Both SN and PE fractions were analyzed by Western blotting.

Bimolecular fluorescence complementation (BiFC) assay

For the BiFC assay, plasmids encoding VNS and SVC were co-transfected into HEK293T cells at a 1:1 molar ratio. Immunofluorescence analysis was performed 24 h post-transfection. Fluorescence images were acquired using a Carl Zeiss LSM 900 confocal microscope and analyzed using Fiji software (ImageJ 1.54i).

Thermal shift assay

Thermal shift assays were performed as previously described⁵¹. Denatured luciferase, with or without YTHDC1 and YTHDC1^{ΔD/E} (1 μM each),

was incubated at 25 °C overnight in a total volume of 9 μL. One microliter of Sypro Orange dye (Invitrogen, Cat# S6650; diluted 1:300 before use) was added to each sample in a 96-well plate. Fluorescence intensity was monitored at a heating rate of 1 °C per minute using a QuantStudio 5 RT-PCR system (Thermo Fisher, USA). The background signal from YTHDC1 was subtracted from the samples.

Fluorescence recovery after photobleaching (FRAP) assay

hnRNPA1 wild-type (WT) and mutants were labeled with Alexa Fluor 647 NHS (Thermo Fisher, Cat# A20006), and YTHDC1 was labeled with Alexa Fluor 488 NHS (Thermo Fisher, Cat# A20000) according to the manufacturer's instructions. For FRAP assays, labeled proteins (50 μM) were mixed with MBP or YTHDC1 in buffer (25 mM Tris, 50 mM NaCl, 10% PEG, pH 7.5) and incubated at 4 °C for 60 min. A 4 μL aliquot of the sample was loaded onto an ultra-thin glass-bottom dish. FRAP experiments were performed using a Carl Zeiss LSM 900 confocal microscope equipped with a ×63 oil objective and a 633-nm laser line. A circular region of approximately 2.5 μm in diameter within droplets (4–6 μm in size) was photobleached for 2.5 seconds at full laser power. Post-bleach images were acquired at 0.5-s intervals. Fluorescence recovery was analyzed using ZEN Lite software. The fluorescence recovery fraction was calculated using the formula: $(I_t - I_{\min}) / (I_t \text{ before bleaching} - I_{\min})$, where I_t is the intensity at a given time point, and $I_{\min}$ is the minimum intensity after bleaching.

Co-Immunoprecipitation (Co-IP) analysis

Co-IP was performed to investigate protein-protein interactions. HEK293 cells were transfected with plasmids encoding HA-tagged hnRNPA1, hnRNPA1^{D262V}, hnRNPA1^{D262N}, CSNK2A1, or Flag-tagged YTHDC1. After 24 h, the cells were lysed in 400 μL of IP lysis buffer (20 mM Tris-HCl, pH 7.4, 150 mM NaCl, 0.5% Triton X-100, 0.5% NP-40, and 10% glycerol) for 30 min on ice. The lysates were then centrifuged at 14,000 × *g* for 30 min at 4 °C, and the resulting supernatants were incubated overnight at 4 °C with Anti-Flag M2 magnetic beads (Sigma, Cat# A8823). Subsequently, the beads were washed, and bound proteins were eluted by denaturation in 2× SDS loading buffer at 95 °C for 10 min. Protein samples were then subjected to Western blot analysis.

Mass spectrometry analysis

HEK293T or SY5Y cells were transfected with either empty vector (control), Flag-tagged YTHDC1, or Flag-tagged YTHDC1^{ΔD/E} or Flag-tagged hnRNPA1 constructs. After transfection, cells were harvested and subjected to immunoprecipitation (IP) using the Anti-Flag M2 magnetic beads (Sigma, Cat# A8823) according to standard procedures. The immunoprecipitated proteins were denatured in SDS buffer (without bromophenol blue), resolved by SDS-PAGE, and analyzed via liquid chromatography-tandem mass spectrometry (LC-MS/MS) at the Biomedicine Public Service Platform in Jiangbei New District, Nanjing, China.

To examine phosphorylation profiles, HEK293T cells were co-transfected with Flag-tagged YTHDC1 along with either an empty vector or siRNA targeting CSNK2A1. Proteins enriched by FLAG-IP were separated by SDS-PAGE, and the band corresponding to Flag-YTHDC1 was excised for in-gel digestion. Phosphopeptide analysis was conducted using an Orbitrap Fusion Lumos mass spectrometer (Thermo Scientific) at the same facility, following established methods⁴². The resulting MS/MS data were processed with Proteome Discoverer software (v2.4, Thermo Fisher Scientific) under the following parameters: database, UniProt human proteome; enzyme, Trypsin/P (up to two missed cleavages allowed); peptide length, at least six amino acids; fixed modification, carbamidomethylation; variable modifications, methionine oxidation and N-terminal acetylation.

In vitro kinase assays

In vitro kinase assays for YTHDC1 were conducted as previously described. Flag-tagged CSNK2A1 proteins, purified from

HEK293T cells, were mixed with recombinant YTHDC1 in kinase reaction buffer (50 mM Tris-HCl, pH 7.0, 10 mM MgCl₂, 1 mM DTT) containing 500 μ M ATP- γ -S (Sigma, Cat# A1388). The reaction mixtures were incubated at 30 °C for 30 min and then terminated by adding 20 μ M EDTA. Samples were denatured at 95 °C for 10 min in 4 $\times$ loading buffers and analyzed by Western blotting.

Detection of amyloid-like protein aggregation in cell lysates

To quantify amyloid-like protein aggregation in cell lysates, cells were lysed in ice-cold RIPA buffer supplemented with protease inhibitors, briefly sonicated, and centrifuged at 12,000 $\times$ g for 15 min at 4 °C to remove debris. The protein concentration of the supernatant was determined by BCA assay. A fresh ThT (Sigma, Cat# 596200) stock solution (5 mM in PBS, filtered through a 0.22 μ m filter) was diluted to 50 μ M in PBS. For each well of a black 96-well plate, 20 μ g of lysate protein was mixed with 180 μ L of 50 μ M ThT and incubated for 30 min at room temperature in the dark. Fluorescence intensity was measured using a multimode plate reader (excitation 440 nm, emission 485 nm). Background fluorescence from lysis buffer alone was subtracted, and net values were normalized to the 20 μ g protein input. Relative aggregation levels were expressed as fold change over control samples, with three independent experiments each performed in technical triplicates.

RNA immunoprecipitation (RIP)

HEK293T cells, either control or subjected to CSNK2A1 knockdown via siRNA, were collected (20 $\times$ 10⁶ cells per IP reaction) and washed twice with ice-cold PBS. The cells were then lysed in ice-cold IP lysis buffer (0.5% sodium deoxycholate, 50 mM Tris-HCl pH 7.5, 150 mM NaCl, 1% NP-40, 0.1% SDS) for 30 min on ice. The lysate was clarified by centrifugation at maximum speed for 10 min at 4 °C to remove debris. The supernatant was incubated with 5 μ g of anti-YTHDC1 antibody (Abcam, Cat# ab122340) or an equivalent amount of rabbit IgG overnight at 4 °C with rotation. RNA-protein-antibody complexes were captured using Dynabeads™ Protein A/G (Thermo Fisher Scientific, Cat# 80106G) and washed five times with NT2 buffer (50 mM Tris-HCl pH 7.5, 150 mM NaCl, 1 mM MgCl₂, 0.05% NP-40). Finally, RNA was extracted by phenol-chloroform purification and quantified for subsequent qRT-PCR analysis.

RNA isolation and quantitative PCR (qPCR)

Total RNA was isolated from cultured cells using Trizol reagent (Vazyme, Cat# R401) following the manufacturer's instructions. RNA samples of equal quantity were then reverse transcribed into cDNA using HiScript III RT SuperMix for qPCR (Vazyme, Cat# R323). Quantitative PCR was carried out on a QuantStudio 5 Real-Time PCR System (Thermo Fisher Scientific, USA) with SYBR Green master mix (Vazyme, Cat# Q712) and specific primer sets. The following primers were used in this study:

Luc-DM. Forward: 5'-GCTCTGCCTCATAGAACTG-3'
Reverse: 5'-TGATGGAATGGAACAACACT-3'

CSNK2A1. Forward: 5'-GCTGCGATATGACCACCACT-3'
Reverse: 5'-GAAGGGGTTGGCACTGAAGA-3'

GAPDH. Forward: 5'-TGGCATTGCCCTCAACGACCAC-3'
Reverse: 5'-CCTCTTGTGCTCTTGTCTGGGCG-3'

Stress granule formation

To assess stress granule assembly, HeLa cells were seeded on glass coverslips and allowed to adhere for 24 h. Cells were then transfected with plasmids encoding GFP, YTHDC1-GFP, hnRNPA1, hnRNPA1^{D262N}, or hnRNPA1^{D262V} using Lipofectamine 3000 (Invitrogen, Cat# L3000015) for 6 h. At 24 h post-transfection, SGs were induced by treatment with

0.5 mM sodium arsenite for 1 h. Following induction, cells were fixed with 4% paraformaldehyde in PBS for 10 min, permeabilized with 0.15% Triton $\times$ -100 for 10 min and blocked with PBS containing 1% BSA and 22.52 mg/mL glycine. For immunostaining, cells were incubated overnight at 4 °C with a mixture of anti-Flag (Sigma-Aldrich, Cat# F1804) and anti-TIAR (Cell Signaling Technology, Cat# 8509S) antibodies. After washing, Alexa Fluor 594-conjugated (Invitrogen, Cat# A-11012) and Alexa Fluor 647-conjugated (Invitrogen, Cat# A-21245) secondary antibodies were applied for signal detection. Coverslips were mounted using ProLong™ Gold Antifade Mountant with DAPI (Invitrogen, Cat# P36966) to visualize nuclei. Images were acquired on a Carl Zeiss LSM 900 confocal microscope and processed with ZEN Lite software. Recruitment efficiency of Flag-tagged hnRNPA1 and its mutants into SGs was quantified as the percentage of transfected cells exhibiting clear co-localization of the protein with TIAR-positive granules.

RT-QulC assay

Samples containing 20 μ M hnRNPA1 or its variants, 5 μ M YTHDC1, and 20 μ M ThT were shaken at 800 rpm at 25 °C in PBS buffer. ThT fluorescence was measured using a microplate reader (BioTek, USA) with excitation at 450 nm and emission at 485 nm, recorded every 5 s. Three replicates were performed for each sample.

Negative-staining transmission electron microscopy (TEM)

For TEM analysis of in vitro aggregation reactions, 4 μ L of the sample was adsorbed onto glow-discharged 300-mesh Formvar/carbon-coated copper grids (Beijing Zhongjingkeyi Technology, China) and stained with 2% (w/v) aqueous uranyl acetate. Excess liquid was removed, and the grids were air-dried. Samples were imaged using a JEM-2800 transmission electron microscope.

Primary neuron extraction and culture

For primary neuron extraction and culture, timed-pregnant C57BL/6 mice at E15 were humanely euthanized via isoflurane anesthesia (4% induction, 2% maintenance). Fetal brains were rapidly extracted under sterile conditions and immersed in ice-cold Hanks' balanced salt solution (Gibco, Cat# 14175095). Cortical tissues were microdissected following careful removal of meninges and vascular structures. After three washes with calcium/magnesium-free PBS (pH 7.4), tissues were mechanically dissociated using fine spring scissors into 1-mm³ fragments. Enzymatic digestion was performed with 1.25% Trypsin-EDTA (Sigma, Cat# T4049) in DMEM/F12 (37 °C, 15 min) with gentle agitation. The digestion was terminated by adding soybean trypsin inhibitor (1 mg/mL, Sigma, Cat# T6522) followed by centrifugation (200 $\times$ g, 5 min). Cell suspensions were filtered through 70- μ m nylon meshes (Corning, Cat# 431751) and seeded at 2.5 $\times$ 10⁵ cells/cm² in neurobasal medium (Gibco, Cat# 21103-049) supplemented with 2% B-27 (Gibco, Cat# 17504-044), 0.5 mM glutamax, and 1% penicillin/streptomycin. Cultures were maintained at 37 °C in a 5% CO₂ humidified incubator with half-medium changes every 72 h. After the cells were wall-affixed, AAV9 at a titer of 1 $\times$ 10⁸ was added and the culture was continued for 7 days.

Synaptic length quantification and sholl analysis

Fluorescence images were acquired with a Carl Zeiss LSM 900 confocal microscope. All image processing and quantitative analyses were performed using Fiji software (ImageJ 1.54i) with customized processing pipelines. For synaptic morphology characterization, individual neurons meeting inclusion criteria were semi-automatically traced using the Simple Neurite Tracer plugin with manual verification, enabling precise quantification of synaptic element lengths through vector-based path measurement. Neuronal complexity was subsequently assessed using a modified Sholl analysis protocol implemented through the Sholl Analysis plugin. Our adaptation incorporated: (1) concentric shell placement at 2 μ m radial increments

from the soma centroid; (2) optimized polynomial curve fitting (4th order) for arborization profile modeling; and (3) automated intersection counting with outlier filtering ($> 3\sigma$ exclusion). Neurites intersection counts and corresponding path lengths at each radius were recorded through batch processing of reconstructed neurites. Raw data of concentric ring intersections and neurites length were compiled and analyzed by GraphPad Prism (version 10.2.0). One-way ANOVA followed by Tukey's post hoc test or two-tailed unpaired Student's *t*-test was used for statistical analysis.

Quantification and statistical analysis

Protein bands from Western blots and fluorescence signals in cells were quantified using ImageJ. Statistical analyses were performed using GraphPad Prism 10. Individual data points are shown in plots and charts. Data are presented as mean $\pm$ standard deviation (SD). Statistical significance was assessed using a two-tailed unpaired Student's *t*-test for comparisons between two groups, and one-way ANOVA with Tukey's post hoc test for multiple comparisons.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

RNA-Seq generated in this article has been deposited in NCBI BioProject database under accession [PRJNA1416556](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1416556); The mass spectrometry proteomics data have been deposited in the ProteomeXchange Consortium via the PRIDE partner repository with dataset identifiers [PXD073870](https://www.ebi.ac.uk/PRIDE/projects/PXD073870) and [PXD073825](https://www.ebi.ac.uk/PRIDE/projects/PXD073825). Source data are provided with this paper. All data are available in the main manuscript or the supplementary material. Requests for materials should be addressed to Y.C. and L.H. Source data are provided with this paper.

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Author contributions

L.H. and Y.C. conceived and supervised the study. L.H., Y.C., and K.R. designed the experiments. K.R. performed most of the experiments. M.C. performed the YTHDC1 immunoprecipitation-mass spectrometry (IP-MS), m⁶A RNA methylation, and site-specific phosphorylation analyses. Q.L. and Y.L. assisted with primary neuron experiments. B.Z. and J.C. assisted with hnRNPA1-related experiments. H.Z. performed data analysis. T.K. and Y.W. participated in protein purification. Z.L. performed the in vitro kinase assays. L.H., K.R., and X.D. wrote the manuscript with input from all authors. All authors reviewed and approved the final manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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